

**SIMATS ENGINEERING**  
**ASSESSMENT TOOL 2: Industry Problem Solving Task**

**COURSE NAME: COMPUTER VISION COURSE CODE: ITA0515 Slot B**

**COURSE OUTCOME COVERED**

**CO4: Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5) Assessment Tool Weightage: 40%**

**QUESTIONS:5 Total Marks 50 Duration :60 Minutes Annexure A**  
**Industry Problem Solving Task**

**Code of Conduct:**

***I V. Thanusri (192472265) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.***

**Signature of the student: V.Thanusri**

**Rubrics for Evaluation**

| Criteria                            | Weightage (%) | Level 4 (Exemplary)                                                          | Level 3 (Proficient)                                 | Level 2 (Developing)                           | Level 1 (Beginning)                               | Marks(100) |
|-------------------------------------|---------------|------------------------------------------------------------------------------|------------------------------------------------------|------------------------------------------------|---------------------------------------------------|------------|
| Understanding of Industry Problem   | 20%           | Comprehensive understanding of real world problem and constraints            | Good understanding with minor gaps in requirements   | Basic understanding; some requirements unclear | Poor understanding; misinterprets problem context |            |
| Application of Engineering Concepts | 25%           | Applies advanced and relevant concepts effectively to solve the problem      | Applies appropriate concepts with minor errors       | Limited application of concepts                | Incorrect or no application of concepts           |            |
| Solution Design                     | 25%           | Innovative, practical, and feasible solution aligned with industry standards | Logical and feasible solution with minor gaps        | Basic solution with limited feasibility        | Unclear or impractical solution                   |            |
| Use of Tools                        | 15%           | Effective use of modern tools, technologies, and real data                   | Appropriate use of tools with correct implementation | Limited or inconsistent use of tools           | Minimal or incorrect use of tools                 |            |
| Analysis                            | 10%           | Thorough analysis with validated results and performance evaluation          | Good analysis with reasonable validation             | Basic analysis with limited validation         | Poor or no analysis                               |            |

|                 |    |                                                                  |                                     |                                    |                                     |  |
|-----------------|----|------------------------------------------------------------------|-------------------------------------|------------------------------------|-------------------------------------|--|
| Professionalism | 5% | Highly professional, well documented, and clearly presented work | Good documentation and presentation | Basic documentation with some gaps | Poor documentation and presentation |  |
|-----------------|----|------------------------------------------------------------------|-------------------------------------|------------------------------------|-------------------------------------|--|

Analyze the given problem by comparing available techniques against the specified constraints and requirements. Evaluate and justify your decision using logical reasoning, clearly indicating how the chosen approach satisfies the conditions.

### 1. Traffic Surveillance Model Selection

A traffic monitoring system must:

- Accuracy  $\geq 90\%$
- Latency  $\leq 80$  ms
- Works under varying lighting

Options:

Method A: 88%, 60 ms

Method B: 91%, 85 ms

Method C: 92%, 75 ms

Compare and evaluate which method satisfies all constraints. Justify logically.

### 2. Drone Vision System Constraint

A drone system requires:

- Lightweight computation
- Real-time detection (<60 ms)
- Accuracy  $\geq 85\%$

Options:

Model A: 83%, 40 ms

Model B: 87%, 65 ms

Model C: 86%, 55 ms

Compare and evaluate the suitable model. Justify using constraints.

### 3. Security System Recognition

A security system must:

- High accuracy (>92%)
- Moderate computation

Options:

Traditional: 85%, low computation

Deep Learning: 94%, high computation

Hybrid: 91%, moderate computation

Compare and evaluate the best approach. Justify logically.

### 4. Motion Tracking in Sports Analytics

Requirements:

- Accurate motion tracking
- Real-time performance

Options:

Optical Flow: accurate, slow

Motion Estimation: fast, less accurate

Compare and evaluate the suitable approach. Justify using constraints.

### 5. Face Recognition System Design

Constraints:

- Accuracy  $\geq 90\%$
- Works in low light
- Moderate computation

Options:

Traditional: 82%

Deep Learning: 93%

Hybrid: 89%

Compare and evaluate the best choice. Justify logically.

## Answers

### 1. Traffic Surveillance Model Selection

Given Requirements

The traffic monitoring system must have:

- Accuracy of at least 90%.
- Latency of 80 ms or less.
- Ability to work under varying lighting conditions.

### Comparison and Evaluation

**Method A:-** has an accuracy of 88% and latency of 60 ms. Its latency satisfies the requirement, but its accuracy is below 90%. Therefore, Method A cannot be selected.

**Method B :-**has an accuracy of 91% and latency of 85 ms. Its accuracy satisfies the requirement, but its latency is higher than the maximum allowed 80 ms. Therefore, Method B cannot be selected.

**Method C :-**has an accuracy of 92% and latency of 75 ms. Both values satisfy the given requirements. It provides higher accuracy than required while maintaining latency below 80 ms.

### Decision

Method C is the best choice. It satisfies the required accuracy and latency constraints.

Therefore, it is the most suitable method for the traffic surveillance system.

### **Drone Vision System Constraint**

Given Requirements

The drone vision system must:

- Use lightweight computation.
- Detect objects in less than 60 ms.
- Achieve accuracy of at least 85%.

### **Comparison and Evaluation**

**Model A:-** has an accuracy of 83% and detection time of 40 ms. Although it is fast and suitable for real-time processing, its accuracy is below the required 85%. Hence, it cannot be selected.

**Model B:-** provides 87% accuracy and takes 65 ms for detection. Its accuracy is sufficient, but the detection time exceeds the 60 ms limit. Therefore, it does not satisfy the real-time requirement.

**Model C:-** provides 86% accuracy with a detection time of 55 ms. Both values satisfy the specified constraints. Its performance is suitable for real-time drone applications.

### **Decision**

Model C is the best choice. It satisfies the minimum accuracy and real-time detection requirements. Therefore, it provides the best balance between accuracy and processing speed.

### **Security System Recognition**

Given Requirements

The security recognition system requires:

- Accuracy greater than 92%.
- Moderate computational requirements.

### **Comparison and Evaluation**

The Traditional method provides only 85% accuracy and uses low computation. Although its computational requirement is good, its accuracy is far below the required level. Therefore, it is not suitable.

The Deep Learning method provides 94% accuracy, which is above the required 92%.

However, it requires high computation. Even though computation is higher than preferred, it provides the required high recognition accuracy.

The Hybrid method provides 91% accuracy with moderate computation. Its computation satisfies the requirement, but its accuracy is below 92%. Therefore, it cannot satisfy all the requirements.

### **Decision**

Deep Learning is the best choice because accuracy is the most important requirement for a security recognition system. It achieves 94% accuracy, which is the only option that exceeds the required accuracy. The higher computational requirement is the main limitation.

### **Motion Tracking in Sports Analytics**

Given Requirements

The sports analytics system requires:

- Accurate motion tracking.
- Real-time performance.

### **Comparison and Evaluation**

**Optical Flow** provides accurate motion tracking and can identify detailed movement. However, it is computationally slower. Since the system requires real-time performance, its slow processing becomes a major limitation.

**Motion Estimation** is faster and can process movement in real time. Its accuracy is lower compared with Optical Flow, but it provides better processing speed.

In sports analytics, tracking must be performed quickly so that the movement of players or objects can be analysed without significant delay. Therefore, processing speed is an important factor.

### **Decision**

Motion Estimation is the suitable approach because it provides fast processing and supports real-time tracking. Although Optical Flow is more accurate, its slow processing does not satisfy the real-time requirement.

## **5. Face Recognition System Design**

### **Given Requirements**

The face recognition system must:

- Achieve accuracy of at least 90%.
- Work effectively in low-light conditions.
- Require moderate computation.

### **Comparison and Evaluation**

The Traditional method provides only 82% accuracy. Although traditional techniques may require less computation, the accuracy is below the required 90%. Therefore, it cannot be selected.

The Deep Learning method provides 93% accuracy, which satisfies the required accuracy. Deep learning techniques can also learn complex facial features and generally provide better performance when lighting conditions change.

However, they usually require higher computational resources.

The Hybrid method provides 89% accuracy with moderate computation. Although its computation requirement is suitable, its accuracy is below the required 90%. Therefore, it does not completely satisfy the requirements.

### **Decision**

Deep Learning is the best choice because it achieves 93% accuracy and provides better robustness under varying lighting conditions. Although its computational requirement may be higher than moderate, it is the only option that clearly satisfies the accuracy requirement and is therefore the most suitable overall.